Cell Tracking Challenge

A reproducible classical computer vision baseline for counting moving and growing bacteria

40	640 x 240	3 to 19

Objective

Estimate the number of bacteria automatically in every frame of the supplied animated GIF. Only the left-half real microscopy image is analyzed; the right-hand laboratory tracking visualization is not used as an answer source.

Approach in one sentence

Highlight locally dark rods by subtracting a lightly denoised image from a broad background estimate, then apply Otsu thresholding, morphological opening, connected components, and area filtering.

Scope and evidence

- All 40 frames are processed by the same parameters; no frame-specific answer is hard-coded.
- Output format is one integer per line, and its length is asserted against GIF metadata.
- No ground truth is supplied, so the report presents sanity checks rather than accuracy claims.

Dataset and Method

The GIF contains 40 consistently sized RGB frames with array shape (240, 640, 3). The algorithm crops columns 0-319 before any image processing, ensuring that the right-half visualization cannot influence detection.

Processing pipeline

- Convert the 320 x 240 microscopy crop to 8-bit grayscale.
- Apply Gaussian denoising with radius 1 pixel.
- Estimate slow background illumination with Gaussian radius 10 pixels.
- Compute background minus denoised intensity so dark rods have positive local contrast.
- Choose a per-frame Otsu threshold, with a minimum response of 12 gray levels.
- Apply a 3 x 3 morphological opening to suppress isolated texture.
- Label 8-connected components and retain areas from 20 through 500 pixels.
- Write the retained component count for the frame.

Parameter	Value	Purpose
Denoise radius	1 px	Suppress pixel-scale texture
Background radius	10 px	Remove slow illumination variation
Threshold	Otsu; floor 12	Adapt automatically per frame
Morphology	3 x 3 opening	Remove tiny isolated responses
Component area	20-500 px	Reject speckles and broad artifacts
Connectivity	8	Keep diagonal rod pixels connected

Why classical CV?

The animation is small, the cells have consistent local appearance, and the requested task is counting. This approach is inspectable, requires no labels or pretrained detector, and makes segmentation errors visible in a binary mask.

Results and Sanity Checks

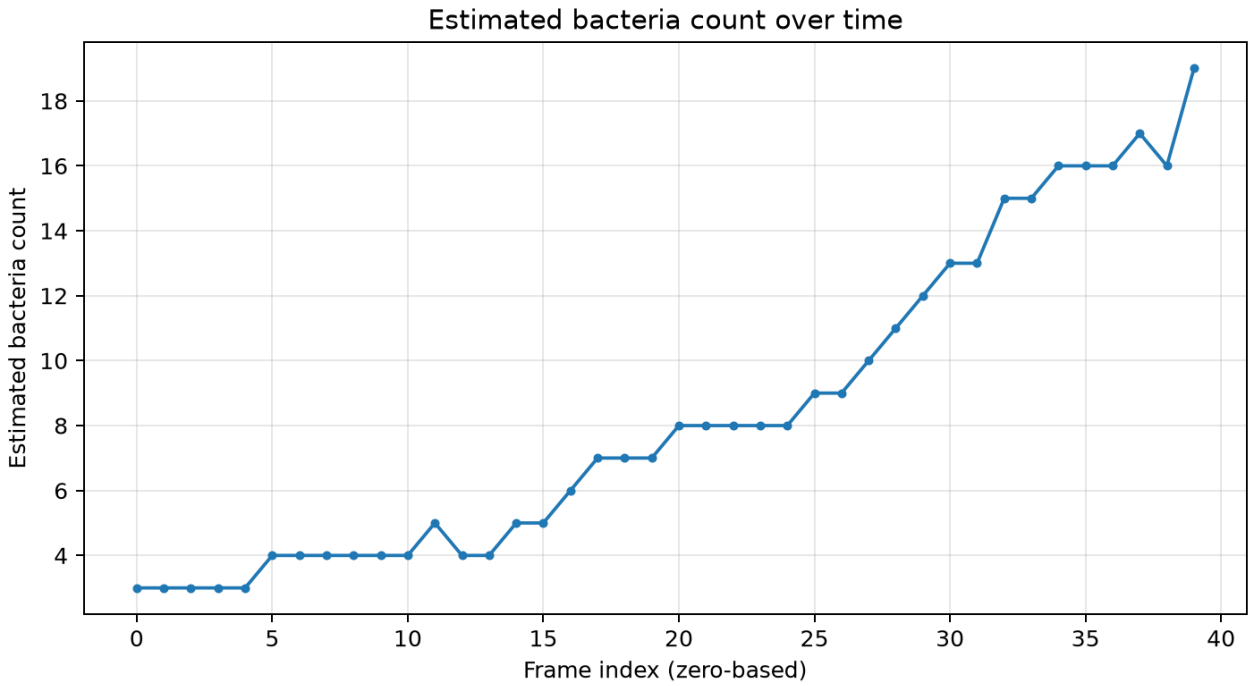


Figure 1. Estimated bacteria count by zero-based frame index. Monotonic growth was not forced; the small downward changes remain visible as segmentation uncertainty.

Frame (zero-based)	Estimated count	Interpretation
0	3	Small initial colony; three dark cores retained
20	8	Mid-sequence growth with separated rods
39	19	Dense final colony; contact risk is highest
All frames	3-19	Overall range; 40 output lines

Observed temporal behavior

The sequence generally increases from 3 to 19. Most adjacent changes are 0 or +1. The largest change is +3 at frame 39; frame 32 changes by +2. Two -1 changes occur, consistent with a marginal component merging or dropping below threshold rather than a manually smoothed output.

These observations make the output plausible, but they do not establish correctness without independent cell annotations.

Qualitative Segmentation Checks

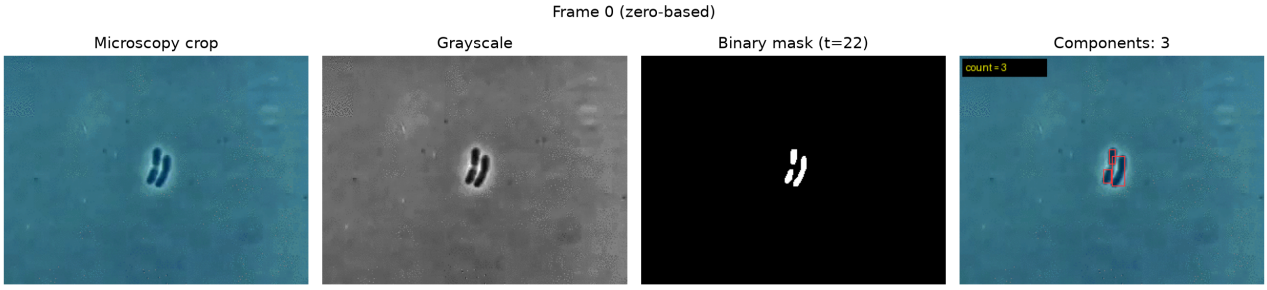


Figure 2. First frame: microscopy crop, grayscale image, binary mask, and retained component boxes. Estimated count = 3.

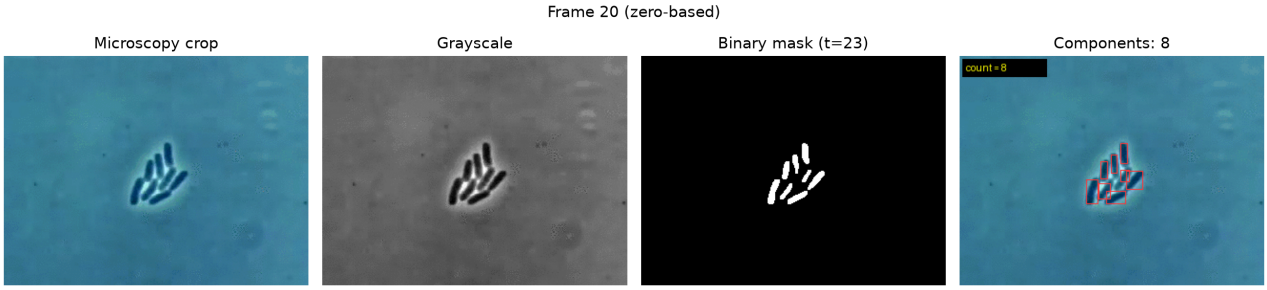


Figure 3. Middle frame: microscopy crop, grayscale image, binary mask, and retained component boxes. Estimated count = 8.

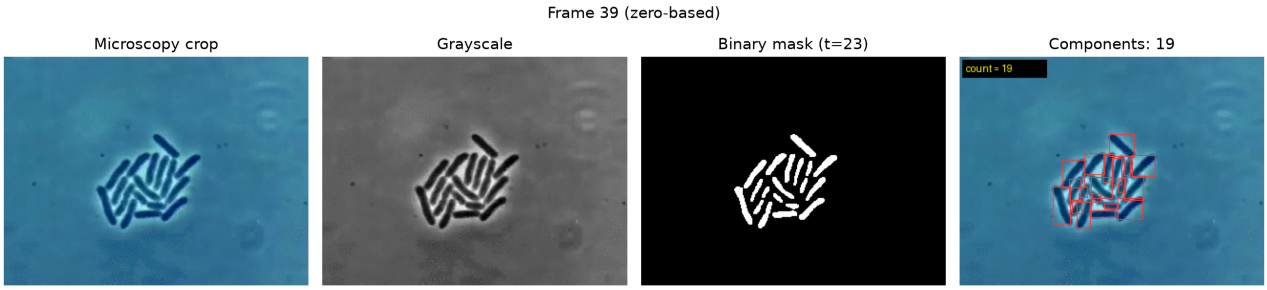


Figure 4. Last frame: microscopy crop, grayscale image, binary mask, and retained component boxes. Estimated count = 19.

Failure Analysis and Limitations

Noise and uneven illumination

GIF compression texture and background gradients can resemble weak dark objects. Local background subtraction, opening, and area filtering remove most of this signal, but a strong artifact could still survive.

Touching cells and dense colonies

Connected components can merge two touching dark cores, causing undercount. The risk rises in late frames. Conversely, a broken response along one rod can form two retained components and cause overcount.

Low contrast and area filtering

A faint cell may fall below Otsu's threshold or below the 20-pixel area cutoff. A large merged group could exceed 500 pixels. These fixed area limits may not transfer to other magnifications.

Why watershed was not added

The inspected dark cores are mostly separated, and the count curve is stable. Without ground truth, marker-controlled watershed would add tuning freedom and could split single rods. It is worth testing only after annotating dense late frames and measuring split errors.

Possible improvements

- Annotate representative early, middle, and dense late frames for objective parameter validation.
- Evaluate watershed only on confirmed merged components.
- Use temporal correspondence to distinguish division from segmentation flicker.
- Extend from per-frame counting to cell identity, lineage, and true tracking.

References

1. Pillow ImageSequence documentation: pillow.readthedocs.io/en/stable/reference/ImageSequence.html
2. Pillow ImageFilter documentation: pillow.readthedocs.io/en/stable/reference/ImageFilter.html
3. NumPy sliding_window_view documentation: numpy.org/doc/stable/reference/generated/numpy.lib.stride_tricks.sliding_window_view.html
4. Matplotlib savefig documentation: matplotlib.org/stable/api/_as_gen/matplotlib.pyplot.savefig.html

The professor's published 2D-case paper was not used to construct this solution.